10.10 Vacuum Bubbles and Tadpoles

LecNote8 builds $Z(J)$ of φ^3 theory from Feynman diagrams and then makes two families of them disappear: those without sources (p. 8: "all the vacuum diagrams being zero") and those with one source (p. 13: every tadpole "is canceled by X"). Page numbers without a LecNote refer to LecNote8. A reader stops and asks:

- Vacuum diagrams are integrals of propagators. How can they be zero, and if the infinities cancel in the normalized ratio (p. 12), why do we need counterterms?
- Why does the field have a vacuum value (p. 9), and why must $\Delta(0)$ be imaginary (p. 11)?
- What goes wrong in LSZ if $\langle 0|\varphi|0\rangle \neq 0$ (p. 10), and how can one X cancel every different tadpole?

In short: the vacuum diagrams add up to $-i\Delta E_0 T$, the phase from the shift ΔE_0 of the vacuum energy. It does not depend on J, so it drops out of every correlation function, and it takes nothing else with it. A tadpole is the field's response to a linear term hidden in φ^3, $\frac{g}{2}\langle\varphi^2\rangle\varphi$, where $\langle\varphi^2\rangle = \Delta(0)/i > 0$ is the vacuum fluctuation of the field at a point. Y cancels it, which moves the expansion point to the true minimum, as LSZ requires; tadpoles cancel as a sum, order by order, because each hangs on a line of zero momentum.
Conventions are Srednicki's: $g_{\mu\nu} = \mathrm{diag}(-,+,+,+)$, $k^2 = -m^2$ on shell, $\tilde{\Delta}(k) = 1/(k^2 + m^2 - i\epsilon)$, $(-\partial^2 + m^2)\Delta(x) = \delta^4(x)$, $\langle 0|T\varphi(x)\varphi(y)|0\rangle = \Delta(x-y)/i$, and $\mathcal{L}_1 = \frac{g}{6}Z_g\varphi^3 + \mathcal{L}_{\rm ct}$ with $\mathcal{L}_{\rm ct} = -\frac{1}{2}(Z_\varphi - 1)\partial\varphi\partial\varphi - \frac{1}{2}(Z_m - 1)m^2\varphi^2 + Y\varphi$. As on p. 10, $Z_g = 1$ at the orders used here. A line is Δ/i, a vertex $ig\int d^4x$, a source $i\int d^4x\, J$, the Y vertex (the X of the pages) $iY\int d^4x$, and each diagram is divided by its symmetry factor S (§10.9). $\langle\varphi^2\rangle$ means $\langle 0|\varphi(x)^2|0\rangle$ in the free theory, and $\mathcal{V}$ is the volume of space (V counts vertices).

What the $\propto$ and $Z(0) = 1$ remove

Since $\frac{1}{i}\delta/\delta J(x)$ acting on $e^{i\int J\varphi}$ brings down $\varphi(x)$, any function of φ in the exponent can be taken out of the integral as the same function of $\delta/i\delta J$, as LecNote5 pp. 9–10 did for one coordinate:
$Z(J) = \exp\bigl[i\int d^4x\,\mathcal{L}_1(\delta/i\delta J(x))\bigr]\int\mathcal{D}\varphi\, e^{i\int(\mathcal{L}_0 + J\varphi)}$. The remaining integral is $Z_0(J) = \exp\bigl[\frac{i}{2}\int J\Delta J\bigr]$ times a J-independent constant: the normalization of $\mathcal{D}\varphi$ and the overlaps of whatever states were prepared at the two ends with the vacuum, which the $(1 - i\epsilon)$ projection leaves behind (Eq. (10.8.13); LecNote5 p. 8). The $\propto$ of LecNote8 p. 2 hides that constant, and $Z_0(0) = 1$ fixes it.

With $\mathcal{L}_1$ switched on, the same projection selects the ground state of the full Hamiltonian. Its energy is no longer the zero chosen for the free theory (LecNote3 p. 12, LecNote5 p. 8); it is shifted by some ΔE_0, and the amplitude to remain in the vacuum for a time T carries $e^{-i\Delta E_0 T}$. In diagrams (LecNote8 p. 8; §10.9 derives the exponentiation, Eq. (10.9.5)), with C_I the connected diagrams and C_0 the sum of those without sources,

$$e^{i\int \mathcal{L}_1(\delta/i\delta J)}Z_0(J) = e^{\sum_I C_I} = e^{C_0}\,e^{iW_1(J)}, \qquad iW_1(J) = \sum_{I\neq\{0\}} C_I. \tag{10.10.1}$$

e^{C_0} is again J-independent, and $Z_1(0)=1$ divides it out: the $\propto$ of pp. 3–8 hides the factor e^{-C_0}. $Z(0)=\langle 0|0\rangle = 1$ says that without a source the vacuum stays the vacuum (Eq. (10.6.13)).

Vacuum bubbles are not zero

A connected vacuum diagram integrates a product of propagators over the positions of its vertices, and the product depends only on differences of positions. One integral is therefore free, and gives the volume of spacetime, $\int d^4x = \mathcal{V}T$. So C_0 is proportional to $\mathcal{V}T$, and comparing with the phase above,

$$C_0 = -i\,\Delta E_0\,T, \qquad \Delta E_0 = \epsilon_0 \mathcal{V}. \tag{10.10.2}$$

The vacuum bubbles compute the shift of the vacuum energy, ϵ_0 being its density. They are not zero; they merely do not depend on J.
Eq. (10.10.2) can be tested where ΔE_0 is known independently. In $0+1$ dimensions the field is the oscillator coordinate $q(t)$ with $L = \frac{1}{2}\dot q^2 - \frac{1}{2}\omega^2 q^2 + \frac{g}{6}q^3$, i.e. $H_1 = -\frac{g}{6}q^3$. There is no $\mathcal{V}$, a line is $G(t)/i = e^{-i\omega|t|}/2\omega$ (Eq. (10.8.1)), and a vertex is $ig\int dt$. With $\int dt\, e^{-in\omega|t|} = 2/in\omega$ (the $i\epsilon$ makes it converge), the two diagrams at $O(g^2)$, the theta (two vertices joined by three lines, $S=12$) and the dumbbell (two loops joined by one line, $S=8$), are

$$\begin{aligned} C_\theta &= \frac{1}{12}(ig)^2\int dt_1 dt_2 \Big[\frac{e^{-i\omega|t_1-t_2|}}{2\omega}\Big]^3 = \frac{ig^2T}{144\,\omega^4},\\ C_{\rm db} &= \frac{1}{8}(ig)^2\int dt_1 dt_2\,\frac{1}{(2\omega)^2}\,\frac{e^{-i\omega|t_1-t_2|}}{2\omega} = \frac{ig^2T}{32\,\omega^4}, \end{aligned} \tag{10.10.3}$$

so that $\Delta E_0 = iC_0/T = -(\frac{1}{144}+\frac{1}{32})g^2/\omega^4 = -11g^2/288\omega^4$.
Rayleigh–Schrödinger perturbation theory gives the same, term by term. First order vanishes, as there is no vacuum diagram with one vertex. At second order, with $q=(a+a^\dagger)/\sqrt{2\omega}$ and $(a+a^\dagger)^3|0\rangle = \sqrt{6}\,|3\rangle + 3\,|1\rangle$,

$$\Delta E_0 = -\sum_n \frac{|\langle n|H_1|0\rangle|^2}{n\omega} = -\frac{g^2}{288\,\omega^4}\Big[\frac{6}{3}+9\Big] = -\frac{g^2}{144\,\omega^4} - \frac{g^2}{32\,\omega^4}. \tag{10.10.4}$$

The theta is the three-quantum intermediate state: its three lines carry three quanta from one vertex to the other. The dumbbell is the one-quantum state: each vertex closes two legs on themselves, leaving the linear piece $\frac{g}{2}\langle q^2\rangle q$ with $\langle q^2\rangle = 1/2\omega$, a constant force $F = g/4\omega$, which lowers the energy by $F^2/2\omega^2 = g^2/32\omega^4$. The dumbbell is our first tadpole.

Correlators are ratios. Because e^{C_0} multiplies everything, it cancels in

$$\langle 0|T\varphi(x_1)\cdots\varphi(x_n)|0\rangle = \frac{1}{i^n}\frac{\delta^n}{\delta J(x_1)\cdots\delta J(x_n)}\frac{e^{\sum_I C_I}}{e^{C_0}}\bigg|_{J=0} = \frac{1}{i^n}\frac{\delta^n e^{iW_1(J)}}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}: \tag{10.10.5}$$

each diagram of the numerator is a part connected to the external points times vacuum bubbles, and the bubbles sum to e^{C_0}.

What the ratio does not do. The bracket on p. 12 reads the normalization as infinities cancelling between numerator and denominator. The bubbles do diverge (the dumbbell is $\propto \langle\varphi^2\rangle^2 \sim \Lambda^4$, see below), and the ratio removes them, but only because they are J-independent. $\langle 0|\varphi|0\rangle = \delta W_1/\delta J|_{J=0}$ is already a ratio and grows like Λ^2; the propagator's loop corrections diverge too. Only counterterms remove those: Y here, Z_φ and Z_m in LecNote12. (The bracket's probability should read $|\langle f|i\rangle|^2/|Z(0)|^2$.)

The tadpole: a linear term hidden in φ^3

At $O(g)$ LecNote8 p. 9 finds the diagram of a line from x ending on a loop,

$$\langle 0|\varphi(x)|0\rangle = \frac{1}{2}\,ig\int d^4y\,\frac{1}{i}\Delta(x-y)\,\frac{1}{i}\Delta(0) + O(g^3). \tag{10.10.6}$$

Nothing forbids it: $\varphi \to -\varphi$ is not a symmetry of the cubic term, so nothing protects $\langle\varphi\rangle = 0$. To see where it comes from, apply Wick's theorem to three free fields at one point, $\varphi^3 = :\varphi^3: + 3\langle\varphi^2\rangle\varphi$, so that

$$\frac{g}{6}\,\varphi^3 = \frac{g}{6}\,:\varphi^3: + \frac{g}{2}\,\langle\varphi^2\rangle\,\varphi. \tag{10.10.7}$$

The normal-ordered part cannot turn the vacuum into one particle, $\langle k|:\varphi^3:|0\rangle = 0$. The second part is a source term $J\varphi$ with the constant $J = \frac{g}{2}\langle\varphi^2\rangle$, and the field responds to a source as $\langle\varphi(x)\rangle = \int d^4y\,\Delta(x-y)J$ (Eq. (10.6.16)). For constant J the integral is $\tilde\Delta(0) = 1/m^2$, so

$$v \equiv \langle 0|\varphi|0\rangle = \frac{g}{2m^2}\,\langle\varphi^2\rangle + O(g^3), \tag{10.10.8}$$

the minimum of $\frac{1}{2}m^2\varphi^2 - \frac{g}{2}\langle\varphi^2\rangle\varphi$. This is Eq. (10.10.6) once $\frac{1}{i}\Delta(0) = \langle\varphi^2\rangle$, shown next. In general a *tadpole* is any part of a diagram that contains no source and hangs on the rest by a single line; p. 9's loop is the simplest one.

Why $\Delta(0)$ is imaginary. Setting $x = x'$ in LecNote7 p. 3's form of the propagator, $\Delta(x-x') = i\int \widetilde{dk}\, e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')-i\omega|t-t'|}$ with $\widetilde{dk} = d^3k/\big((2\pi)^3 2\omega\big)$, gives

$$\frac{1}{i}\,\Delta(0) = \int \frac{d^3k}{(2\pi)^3\,2\omega} = \langle 0|\varphi(x)^2|0\rangle > 0. \tag{10.10.9}$$

$\langle\varphi^2\rangle$ is the squared norm of $\varphi(x)|0\rangle$, the mean-square vacuum fluctuation of the field at one point, so $\Delta(0)$ is i times a positive number. This settles p. 11:

$$Y = \tfrac{1}{2}\,ig\,\Delta(0) = -\tfrac{g}{2}\,\langle\varphi^2\rangle \tag{10.10.10}$$

is real, as it must be for a real $\mathcal{L}$, and $Y\varphi$ is exactly minus the linear term in Eq. (10.10.7).

Why it grows like Λ^2. Every mode adds $1/2\omega$ to Eq. (10.10.9), so it grows like $\int k\,dk$; the pole at $k^2 = -m^2$ mentioned on p. 11 plays no role, since Eq. (10.10.9) has none. (So did LecNote3's zero-point energy, p. 12 there, each mode adding $\omega/2$; there Ω_0 cancelled it, here $Y\varphi$ does.) The coefficient of Λ^2 depends on the cutoff. A sharp cutoff $|\mathbf{k}| < \Lambda$ gives $\langle\varphi^2\rangle = \Lambda^2/8\pi^2$ at $m = 0$, and picks a frame. The regulator of LecNote8 p. 11 multiplies $\tilde\Delta$ by $[\Lambda^2/(k^2+\Lambda^2-i\epsilon)]^2$, which is ≈ 1 for $|k^2| \ll \Lambda^2$, falls like Λ^4/k^4 beyond, and depends only on k^2, so Lorentz invariance is kept. Its poles lie in the same quadrants of the k^0 plane as those of $\tilde\Delta$, so the Wick rotation $k^0 = ik^0_E$ is allowed (Eq. (10.8.14)), and with $u = k_E^2$

$$\Delta_\Lambda(0) = \frac{i}{16\pi^2}\int_0^\infty du\,\frac{u}{u+m^2}\,\frac{\Lambda^4}{(u+\Lambda^2)^2} = \frac{i}{16\pi^2}\Big[\Lambda^2 + m^2 - m^2\ln\frac{\Lambda^2}{m^2} + \dots\Big], \tag{10.10.11}$$

exactly $i\Lambda^2/16\pi^2$ at $m = 0$, half the sharp-cutoff value: LecNote8 p. 12's $\Delta(0) \simeq i\Lambda^2/16\pi^2$, and $Y \simeq -g\Lambda^2/32\pi^2$. That coefficient is not physics; $\langle 0|\varphi|0\rangle = 0$ is, and it holds at every Λ because Y absorbs whatever $\langle\varphi^2\rangle$ is. (LecNote12 returns to regulators.)

Why LSZ needs Y

LSZ (LecNote4) uses $a_1^\dagger(\pm\infty)|0\rangle$ as a one-particle state. Its vacuum component, $\langle 0|a_1^\dagger(t)|0\rangle = (2\pi)^3\,m\,v\,f_1(\mathbf{0})\,e^{-imt}$ (Eq. (10.6.4)), has constant modulus. It never decays, so for $v \neq 0$ the in and out states keep a piece of vacuum and the derivation fails (§10.6). LecNote4 p. 7 removed v by the shift $\varphi \to \varphi - v$; Y is that shift, made inside $\mathcal{L}$. At this order the field sees the potential $\frac{1}{2}m^2\varphi^2 - \frac{g}{2}\langle\varphi^2\rangle\varphi$, with its minimum at the v of Eq. (10.10.8). Adding $Y\varphi$ with $Y = -m^2v = -\frac{g}{2}\langle\varphi^2\rangle$, Eq. (10.10.10), removes the linear term, and $\varphi = 0$ becomes the minimum we expand about. Without Y the tadpoles would also act on lines: each one hung on a line inserts igv (the blob of Eq. (10.10.6) times the vertex ig), and the chain $\frac{1}{i}\tilde\Delta\sum_n(gv\tilde\Delta)^n = \frac{1}{i}(k^2+m^2-gv)^{-1}$ has mass^2 $m^2 - gv = V''(v)$ for

$V = \frac{1}{2}m^2\varphi^2 - \frac{g}{6}\varphi^3$: the curvature at the shifted point, not the m^2 of $\mathcal{L}_0$ (other loops shift the pole further).

All tadpoles cancel

Y is one number per order, so it cannot cancel each tadpole separately; LecNote8 p. 13's "canceled by X" is a statement about sums. A tadpole hangs on a single line, and no momentum can flow through that line, since the tadpole has no source to supply it. In position space the blob at the end of the line is the same at every point, and $\int d^4z\,\Delta(y-z) = 1/m^2$ for every y. So a tadpole contributes a constant, whatever it hangs on. Fix the rest of a diagram and the vertex y where something hangs, and sum over everything that can hang there at a given order, X included: the sum is that order of $\langle 0|\varphi|0\rangle$, which Y makes zero (p. 12). Diagrams with tadpoles therefore cancel in sets, whatever else they contain.

The simplest set is p. 13's, with two sources at $O(g^2)$: the tadpole on a line, $S = 4$ (Example C of §10.9: the loop and the left–right reflection), and the X on a line, $S = 2$ (the reflection only). With $K = \int iJ(x_1)\,iJ(x_2)\,\frac{1}{i}\Delta(x_1-y)\frac{1}{i}\Delta(x_2-y)\frac{1}{i}\Delta(y-z)$, integrated over all four points,

$$\frac{1}{4}(ig)^2\,\frac{1}{i}\Delta(0)\,K + \frac{1}{2}(ig)(iY)\,K = \frac{1}{2}\,ig\left[\frac{1}{2}(ig)\frac{1}{i}\Delta(0) + iY\right]K. \qquad (10.10.12)$$

The bracket is p. 10's and vanishes for $Y = \frac{1}{2}ig\Delta(0)$. The two symmetry factors differ by the $\frac{1}{2}$ of the tadpole's loop.

Beyond $O(g)$, $Y = Y_1 g + Y_3 g^3 + \ldots$, and at each order the new coefficient is fixed so that all one-source diagrams of that order, including those with lower-order X's (the Yg^2 and Y^2g diagrams of p. 10), add to zero. Those that contain a smaller tadpole or an X cancel among themselves by the argument above, so Y_3g^3 cancels the $O(g^3)$ one-source diagrams with no tadpole below the source line. Y has no even powers. The kinetic and mass terms, counterterms included, are even in φ, and φ^3, φ are odd, so $\varphi \to -\varphi$ turns the theory at (g, Y) into the theory at $(-g, -Y)$ and maps each condition that fixes a counterterm, $\langle 0|\varphi|0\rangle = 0$ among them, into itself. Hence $Y(-g) = -Y(g)$, and the correction to p. 10's Y is $O(g^3)$, not $O(g^2)$. In zero dimensions, with weight $\exp(-\frac{1}{2}\varphi^2 + \frac{g}{6}\varphi^3 + Y\varphi)$ and every line equal to 1, $\langle\varphi\rangle = 0$ gives $Y = -\frac{g}{2} - \frac{g^3}{4} - \frac{5g^5}{8} - \ldots$; the $-\frac{1}{4}$ is minus p. 9's loop with a bubble inserted ($S = 4$).

Summary

- The $\propto$ signs and $Z(0) = 1$ remove J-independent factors only: the overlaps left by the $i\epsilon$ projection, and the bubbles $e^{C_0} = e^{-i\Delta E_0 T}$ (for the cubic oscillator, $\Delta E_0 = -11g^2/288\omega^4$). UV divergences of connected diagrams survive the ratio; counterterms remove them.

- A tadpole is the response to the linear term $\frac{g}{2}\langle\varphi^2\rangle\varphi$ in $\frac{g}{6}\varphi^3$; $\Delta(0) = i\langle\varphi^2\rangle \sim i\Lambda^2$, and $Y = -\frac{g}{2}\langle\varphi^2\rangle$ is real.
- Y puts the expansion point at the true minimum ($v = 0$), as LSZ needs. Tadpoles cancel as a sum, order by order; Y is odd in g.